Elements of AIML Lab
Experiment 01 (After-submission Reference): WEKA for Regression

Tofik Ali

March 1, 2026

Contents

1 How to use this reference **1**

2 Quick recap: what regression means **1**
2.1 Key terms (minimum) 2

3 Worked example in WEKA (end-to-end) **2**
3.1 Dataset choice (example) 2
3.2 Step 1: load data and set class attribute 2
3.3 Step 2: minimal preprocessing (recommended) 2
3.4 Step 3: train and evaluate two regressors 2
3.5 Step 4: interpret the metrics correctly 3
3.6 Step 5: optional error visualization (useful) 3

4 What a strong submission looks like (cross-check) **3**
4.1 Screenshots 3
4.2 Report structure (1–2 pages) 3

5 Troubleshooting (common issues and fixes) **3**

1 How to use this reference

This document is meant to be shared **after you submit** Experiment 01. Use it to cross-check:

- whether you selected the correct **class attribute** (numeric target),
- whether you evaluated correctly (CV or hold-out, not training-set),
- whether you interpreted MAE/RMSE in the **units of the target**.

Numbers in WEKA depend on the dataset and on random seeds/parameters, so your exact values may differ.

2 Quick recap: what regression means

Regression is **supervised learning** where the target is **numeric** (continuous), e.g., price, time, temperature. In WEKA, regression is done in the **Classify** tab using a **regressor** (not a

classifier).

2.1 Key terms (minimum)

- **Model**: a function that maps features x to a numeric prediction $\hat{y}$.
- **MAE**: average absolute error (same units as y).
- **RMSE**: square-root of average squared error (same units as y; punishes large errors).
- **Cross-validation (CV)**: repeat train/test on folds and average to get a stable estimate.
- **Overfitting**: model performs well on training data but poorly on unseen data.

3 Worked example in WEKA (end-to-end)

3.1 Dataset choice (example)

If you do not have a dataset, WEKA often ships with sample datasets under a `data/` folder. For regression, a common example is a housing dataset where the target is a numeric value.

3.2 Step 1: load data and set class attribute

1. Open **WEKA** → **Explorer**.
2. **Preprocess** tab → **Open file...** → load `.arff` or `.csv`.
3. In the right panel, set **Class** to the **target** column (the value to predict).

Cross-check: the class attribute must be **numeric**. If it is nominal, you are doing classification.

3.3 Step 2: minimal preprocessing (recommended)

Typical safe steps:

- Missing values → `Filter` → `unsupervised/attribute/ReplaceMissingValues`.
- Remove obvious IDs (e.g., `CustomerID`) that do not represent a measurable feature.

Record any filter you apply (preprocessing is part of the final pipeline).

3.4 Step 3: train and evaluate two regressors

Model A: Linear Regression.

1. Go to **Classify**.
2. **Choose** → `functions/LinearRegression`.
3. **Test options** → choose **Cross-validation** (10 folds) *or* **Percentage split** (e.g., 70/30).
4. Click **Start**.

In the output, note MAE and RMSE (and correlation coefficient if shown).

Model B: Random Forest (a strong baseline).

1. **Choose** → `trees/RandomForest`.
2. Keep the **same** test option as Model A (fair comparison).
3. Click **Start**.

3.5 Step 4: interpret the metrics correctly

Write metric interpretations in your dataset's units. Example:

- If the target is `Price (INR)` and RMSE = 5000, then typical errors are about **5000 INR**.
- If RMSE is much larger than MAE, your model may be making a few very large mistakes (outliers / hard cases).

3.6 Step 5: optional error visualization (useful)

WEKA can visualize prediction errors for some models. If available:

- in the output list, right-click the result → **Visualize classifier errors**.

This helps you explain *where* the model fails (certain ranges of the target, or certain feature regions).

4 What a strong submission looks like (cross-check)

4.1 Screenshots

- Preprocess tab: dataset loaded + correct **class attribute** selected.
- Classify tab: regressor selected + test option selected.
- Output: metrics (MAE/RMSE) for at least two regressors.

4.2 Report structure (1–2 pages)

1. Dataset: name/source, target definition, number of rows/columns.
2. Preprocessing: missing value handling, removed columns, any transformations.
3. Models: Model A and Model B with evaluation method.
4. Results table: MAE/RMSE (+ any other metrics you used).
5. Observations: 3–5 clear points explaining which model is better and why.

5 Troubleshooting (common issues and fixes)

- **Class attribute is wrong:** re-check the **Class** selection in Preprocess.
- **Evaluating on training set:** in Test options, avoid **Use training set**.
- **Target accidentally nominal:** if the target is stored as text/categorical, convert it to numeric (or choose a proper numeric target).

- **Very poor error values:** check for outliers, wrong units, or missing values not handled.
- **Unfair comparison:** use the same evaluation method for all models.

Sources: Provost and Fawcett, *Data Science for Business.*;McKinney, *Python for Data Analysis.*;WEKA Documentation (Waikato Environment for Knowledge Analysis), accessed 2026-02-28.;Matplotlib and Seaborn documentation, accessed 2026-02-28.